

ROHIT MAHALI

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Education

B.Tech + M.Tech (Integrated), Computer Science & Engineering

2022 – 2027

Central University of Jharkhand

Spec: Machine Learning & Data Science

Technical Skills

AI/ML: PyTorch, NLP, ASR, RAG Systems, Stylometry, Model Evaluation, Representation Learning

Research Focus: Experimental Design, Benchmarking, Error Analysis, Low-Resource NLP, Robustness

Programming: Python, C++, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Systems & Tools: FastAPI, Streamlit, ETL Pipelines, Multiprocessing, Pinecone, GCP, Git

Research Experience

Indigenous Language Development Intern

Remote

IndicWiki Project, IIIT Hyderabad

- Conducted data curation and validation for multilingual, low-resource corpora used in downstream NLP research.
- Designed preprocessing pipelines to improve dataset consistency across Wikipedia sister projects.
- Performed schema alignment and error analysis to enhance cross-lingual dataset reliability.

Projects & Research

1. Dia-RAG: Dialect-Aware Retrieval Architecture (Ongoing Research)

<https://github.com/rohitmahali01/Dia-Rag>

- Proposed a dialect-aware RAG framework and Santali-CultureQA/CSS benchmark to measure hallucinations in low-resource QA.
- **Outcome:** Provides an evaluation-driven blueprint for preserving dialectal nuance and reducing culturally incorrect answers.

2. Catch-AI: Detection of AI-Generated vs Human Text

<https://github.com/rohitmahali01/Catch-Ai>

- Developed a hybrid detection system using stylometric, linguistic, lexical-diversity, and token-distribution features.
- **Outcome:** Enables stronger screening and robustness analysis for distinguishing AI-generated and human-authored text.

3. Drought Prediction using LSTM (Collaborative Project)

- Supported LSTM-based long-term drought forecasting using historical SPI-12 climate data and Python ML workflows.
- **Outcome:** Helps anticipate drought risk earlier, supporting climate monitoring, planning, and data-driven mitigation decisions.

4. Santali Language Suite (ASR + Data Processing)

<https://github.com/rohitmahali01/Santali-Tokenizer> — <https://github.com/rohitmahali01/Whisper-Santali-Asr>

- Built a Whisper-based ASR pipeline with preprocessing for noise handling, text-audio alignment, and WER evaluation.
- **Outcome:** Improves speech technology access for Santali by strengthening low-resource dataset quality and ASR evaluation.

5. Smart Tax Platform (AI Document Analysis)

<https://github.com/rohitmahali01/Smart-Tax-Platform-with-AI-Document-Analysis>

- Designed a document understanding pipeline combining rule-based extraction and LLM-based analysis for noisy financial PDFs.
- **Outcome:** Converts unstructured salary documents into usable insights for faster tax-regime comparison.

Hackathons

LIT-RAG — Bajaj Finserv HackRx 6.0 Semifinalist (Top 38/9000)

<https://github.com/rohitmahali01/LIT-RAG>

- Led a team of 5 to build a high-speed RAG pipeline with Pinecone indexing and Python multiprocessing.
- **Outcome:** Achieved sub-15s document analysis latency, improving retrieval speed for real-time question answering.